Topic categories: 17. Power Electronics for Transportation Electrification; 01. Power Devices (Si / Wide Band Gap) and Applications

Building Block Design for Stacked Polyphase Bridge Converters for High-Voltage Traction Drives

***Abstract*—**In a stacked polyphase bridge (SPB) converter the battery voltage is shared by series-connected three-phase submodules, so the transistor voltage class becomes a design variable. This digest presents the design of the phase-leg building block of a 1200 V, 300 kW SPB traction inverter. Because the phase current does not depend on the number of submodules, 100 V GaN transistors give the lowest loss, and the turn-off overshoot at a realistic commutation-loop inductance sets 16 submodules of 75 V. Four EPC2361 per switch, driven in two identical pairs by one dual-channel isolated driver each, keep the junction temperature at 69 °C. The leg is built as a double-pulse-test (DPT) board with a coaxial shunt in every cell and as a load-test board without shunts. Q3D extraction of the routed boards gives 0.41 nH per cell without and 0.65 nH with the shunt, with negligible coupling between the cells, and gate loops that are equal for the four paralleled devices but reach 20.6 nH. A double-pulse simulation with the complete extracted parasitic network confirms current sharing among the four cells within 3 %; the DPT board needs a slower turn-on and a −1.9 V off-state to stay at 93 V. Double-pulse measurements on the DPT board will be added in the final paper.

I. Introduction

High-voltage traction batteries push two-level inverters to 1200 V transistors and large voltage steps at the machine terminals. The stacked polyphase bridge (SPB) converter [1], [2] connects N three-phase bridges in series on the dc side, each feeding one winding set of a multiphase machine. Every submodule blocks only V_{bus}/N, which allows low-voltage transistors and reduces the voltage step and the dc-link capacitance [3], [4]. Control, dc-link stability and machine design of the SPB have been studied [2]–[6], and a GaN-based SPB was found competitive with a three-level flying-capacitor inverter [7].

The submodule itself has received little attention. At 1200 V and 300 kW each leg conducts 228 A_{rms} at 75 V, so several GaN transistors must be paralleled, and paralleled low-voltage GaN stages are limited by turn-off overshoot [8], uneven dynamic current sharing [9], [10] and gate disturbances. Designs with up to eight devices per switch [11], [12] start from a given bus voltage. In an SPB the submodule voltage is free, and it is coupled to the device count and the layout through the loop inductance. This work contributes: (i) a selection of the device voltage class and the submodule voltage that includes the loop inductance; (ii) a gate-drive arrangement for four paralleled GaN transistors per switch based on a current-sharing sensitivity analysis; (iii) a four-cell leg board in a test and an operating version, verified by Q3D extraction and by a double-pulse simulation with the complete extracted parasitic network.

II. Device and Submodule Voltage

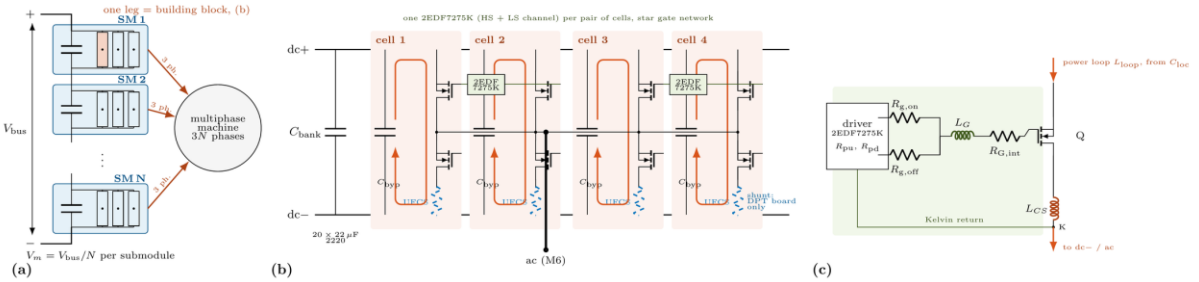


Fig. 1: (a) SPB inverter with N submodules (SM) of $V_m = V_{bus}/N$. (b) Building block, one phase leg: four commutation cells (QH, QL: EPC2361) with bypass capacitors C_{byp}, a common bank C_{bank} and one ac output; one dual-channel driver per pair of cells; shunt (dashed) on the DPT board only. (c) Gate circuit with the common-source inductance L_{CS} up to the Kelvin pad K.

For N submodules sharing the power equally (Fig. 1a), the inverter power is

$$P = 3N\,P_{\text{leg}} = \frac{3M\,V_{\text{bus}}}{2\sqrt{2}}\,I_{\text{ph}}\cos\varphi \tag{1}$$

so the phase current is independent of N: with $M = 1.15$ and $\cos\varphi = 0.9$, 300 kW at 1200 V require 228 A_{rms} (322 A peak) in every leg. Since every submodule carries the full current, the conduction loss scales with $N\,I^2 R_{\text{DS(on)}}/n$ and the output-charge loss with $N\,n\,f_{\text{sw}} C_{\text{oss,tr}} V_{\text{m}}^2$ for n devices per switch. Optimizing n gives

$$P_{\text{min}} = 6\,I\,V_{\text{bus}}\sqrt{f_{\text{sw}} R_{\text{DS(on)}} C_{\text{oss,tr}}} \tag{2}$$

in which the voltage class cancels to first order. Ten GaN transistors of the 100–650 V classes are therefore ranked by the time constant $\tau = R_{\text{DS(on)}} C_{\text{oss,tr}}$ and, since conduction dominates, by the on-resistance per volt of blocking capability (Fig. 2). The 100 V class leads in both; EPC2361 (0.75 mΩ) reaches 7.5 µΩ/V, three to thirty times below the higher-voltage parts. The submodule voltage then follows from the overshoot at turn-off,

$$V_m + L_{\text{loop}}\,di/dt \le 90\ \text{V} \tag{3}$$

A vertical loop over a 0.1 mm dielectric reaches about 0.5 nH per cell, whereas a measurement shunt raises it to about 1 nH. With the turn-on rate of 33 A/ns at 80 A per device, (3) allows 73 V at 0.5 nH and 57 V at 1 nH. An even N is required for symmetric winding sets, so $N = 16$ and $V_{\text{m}} = 75$ V are selected; the operating hardware must therefore stay close to 0.5 nH.

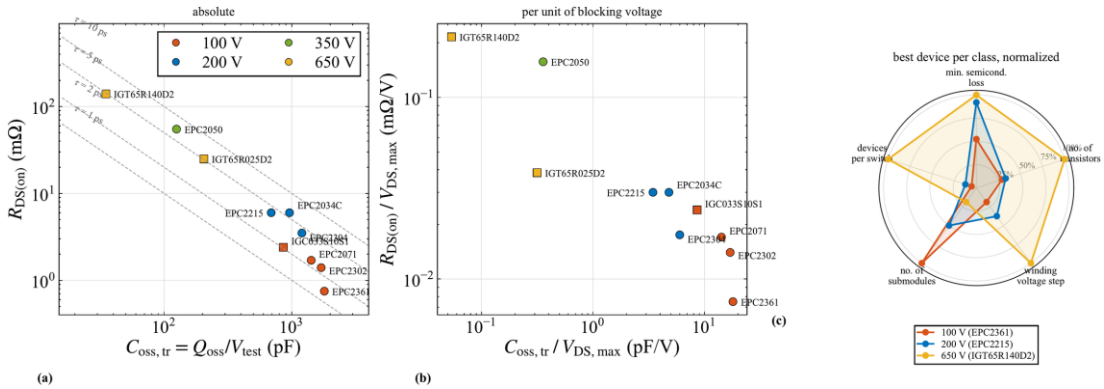


Fig. 2: Device selection (datasheet values). (a) $R_{\text{DS(on)}}$ against $C_{\text{oss,tr}}$ with lines of constant τ; (b) both per volt of blocking capability; (c) SPBs from the best device of each class, normalized: minimum loss (2), devices per switch, submodules, voltage step and transistor count.

III. Devices per Switch, Gate Drive and DC Link

The leg loss over a fundamental period is computed from LTspice switching energies at 0.5 and 1 nH, with the junction temperature and $R_{\text{DS(on)}}$ solved together for the thermal path of the load-test hardware (fan heatsink 0.65 K/W, 7.6 K/W from junction to heatsink per device). One device per switch runs away thermally and two reach 141 °C (Fig. 3). Three devices exceed 90 V. Four devices dissipate 18.0 W per leg at 69 °C and 40 °C ambient with 83 V peak at 0.5 nH, but 93 V at 1 nH; four per switch are chosen.

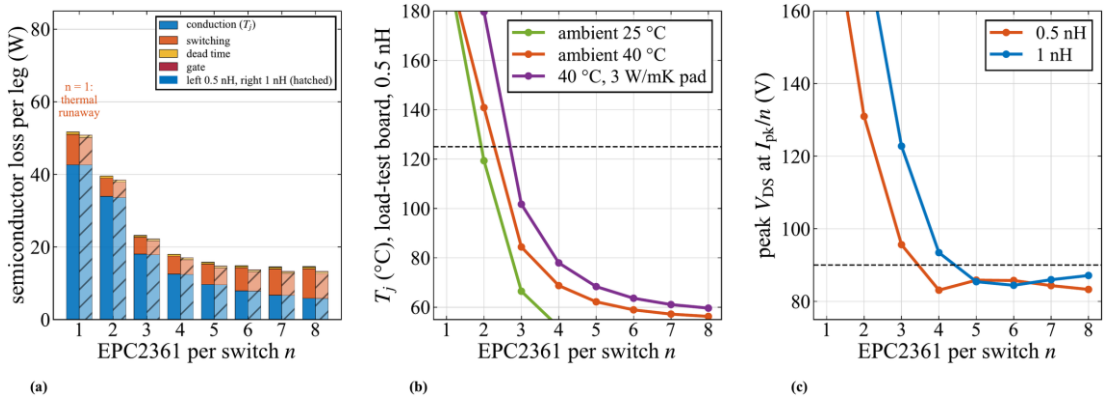


Fig. 3: n EPC2361 per switch at 228 A_{rms} and 50 kHz. (a) Semiconductor loss per leg, 0.5 nH (left) and 1 nH (right, hatched); (b) junction temperature on the load-test heatsink; (c) peak v_{DS} at the peak current per device.

Table I: Switching-energy imbalance of two paralleled EPC2361 (LTspice, 75 V, 120 A)

Asymmetry	**Imbalance**	**Layout measure**
Power loop, L_D +0.3 nH	2 %	symmetric cells
Gate loop, L_G 2 vs. 8 nH	19 %	star gate network
No Kelvin, L_{CS} +0.2 nH	29 %	Kelvin return per device
Threshold, V_{th} +0.3 V	7 %	one channel per pair

Dynamic sharing is dominated by the gate loop and the common-source path, not by the power loop (Table I). One channel for all four devices of a switch is excluded, since the slower gate charging raises the turn-on energy by 31 % and the sink current exceeds the 8 A rating. One 2EDF7275K (isolated high- and low-side channel) is therefore placed between two cells rotated by 180° about it and drives both gates through a star with Kelvin returns (Fig. 1b), so both gate branches are identical; a Zener split of its supply gives +5.1 V and −1.9 V against the gate bump of the off device, inside the −4 V gate limit. The three legs of a submodule share one dc bus, so only the switching-frequency ripple remains: 3 % ripple requires 127 µF effective per leg, and with only 29 % of the nominal value left at 75 V, twenty 22 µF 2220 MLCCs are placed on each leg board, plus ten 1 µF 0805 bypass capacitors per cell.

IV. Leg Board in Two Versions

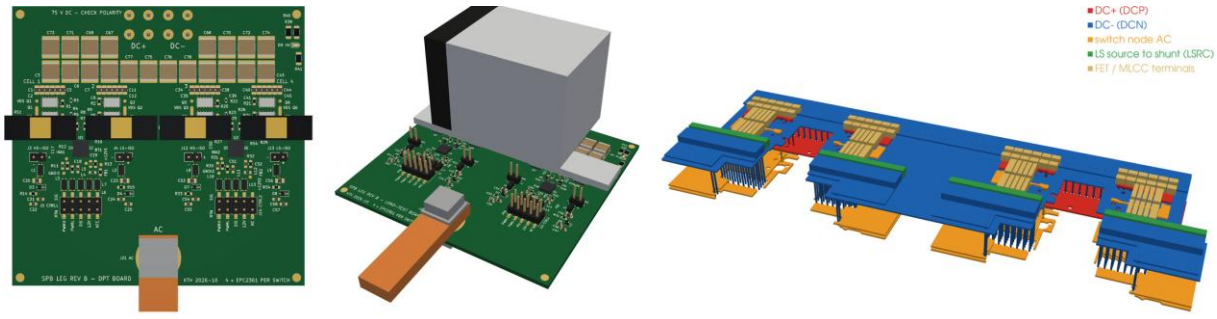


Fig. 4: Leg board, four EPC2361 per switch (89.5 × 85.2 mm, four layers). (a) DPT board with a coaxial shunt per cell; (b) load-test board with fan heatsink; (c) Q3D model of the cells, generated from the routed copper (red dc+, blue dc−, orange switch node).

The four-layer board (2 oz, 0.1 mm dielectric between L1 and L2) carries the cells as two identical pairs, each with its driver in the centre; every commutation loop closes vertically on the dc− plane directly below (Fig. 4). Vias removed for clearance are removed together with their mirror image. The driver inputs, with one common-mode choke per signal pair and a ferrite bead in the 12 V feed, keep 4.0 mm (outer) and 3.0 mm (inner layers) from all power copper, the IPC-2221B spacing for 1200 V, so the layout is ready for a reinforced isolated driver; the 2EDF7275K used here provides functional isolation for the 75 V tests. Measuring every cell current requires a shunt in each low-side source, which lengthens the loop and blocks the heatsink, so the same design is built twice: the DPT board carries a 1 mΩ coaxial shunt per cell, MMCX jacks for v_{DS} and spring-probe pads for v_{GS} of every device; the load-test board

connects the sources through blind vias to the dc− plane and carries the heatsink. Q3D extraction with all four cells switching gives 0.41 nH per cell on the load-test board (0.35 nH copper, 0.06 nH capacitor ESL), equal in the four cells within 1.6 %, and 0.65 nH on the DPT board, where the shunt and its connection add 0.24 nH. The cells hardly couple (at most 21 pH, $k \leq 0.034$). A second model adds the gate network of one driver block: gate and Kelvin return run on adjacent layers to the driver, and a device sees 20.6 nH (high side) and 12.3 nH (low side) when its pair is driven, equal within 0.3 % for all four paralleled devices, with at most 9 pH coupling from the commutation loops. This is about ten times the 1.7 nH of the lumped design, because the driver sits between two cells behind the isolation barrier. Three load-test boards form a level on a laminated PCB busbar with 1.46 nH between neighbouring legs.

V. Validation With the Extracted Parasitics

To check the lumped design against the routed copper, the extracted network was inserted into a double-pulse circuit: the 100 MHz partial-inductance matrix of cells, bypass pads, shunt lands and switch-node copper as one inductor per terminal with every mutual coupling (29 inductors and 400 couplings for the load-test board), the dc resistance matrix including its mutual terms as behavioural sources, eight EPC2361 models, the bypass capacitors with dc-bias capacitance and ESL, the shunts, the bank and two drivers with star gates and Kelvin returns. The load current is switched at the peak phase current (Fig. 5).

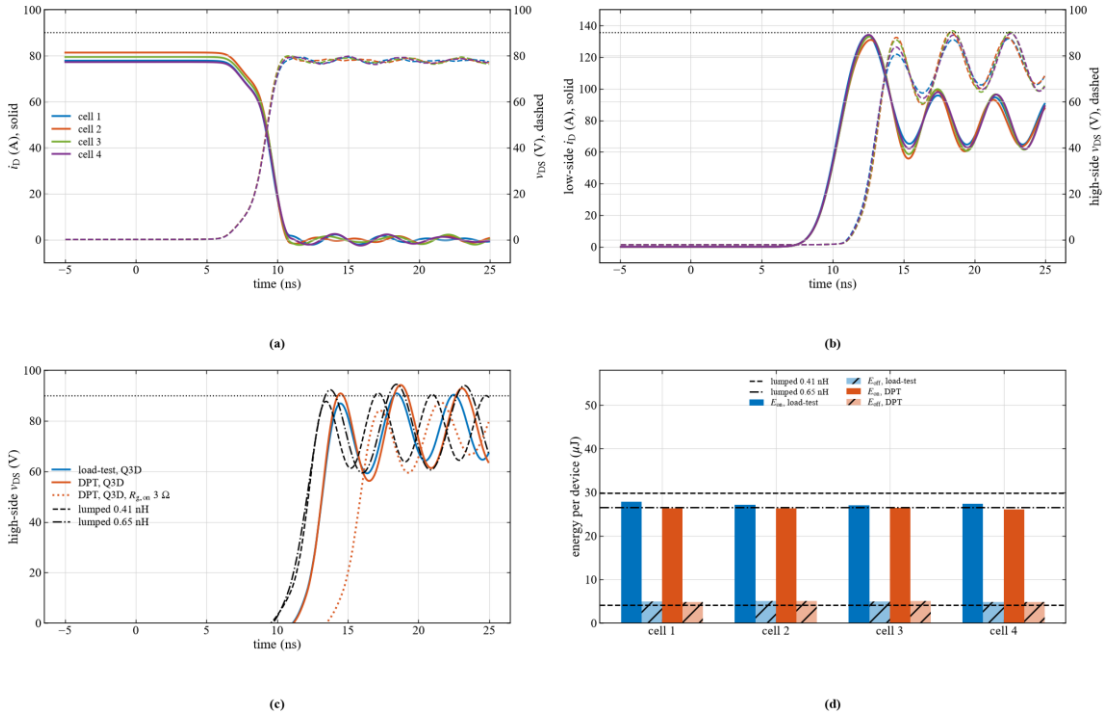


Fig. 5: Double-pulse simulation with the Q3D-extracted copper, 75 V, 316 A (≈ 80 A per device), $R_{g,on} = 2\ \Omega$. (a) Low-side turn-off and (b) turn-on on the load-test board, four cells: drain current (solid), v_{DS} (dashed). (c) High-side v_{DS} at the low-side turn-on vs. the lumped model. (d) Switching energy per device in each cell; lines: lumped model.

The four cells carry 77–81 A at turn-off and their turn-on energies differ by 2 %, so the symmetric layout and the per-pair drive leave none of the mechanisms of Table I active. The low-side turn-off is uncritical (81 V). The critical event is the low-side turn-on, at 46 A/ns: the high-side device peaks at 91 V on the load-test board, in agreement with the lumped model at the extracted loop (91 V). Turn-on energies agree with the lumped model within 10 %. With $R_{g,on} = 3\ \Omega$ the turn-on slows to 34 A/ns and the peak drops to 86 V for a 20 % higher turn-on energy (below 1 W per leg). The gate bump of 1.2 V with a 0 V off-state confirms the need for the negative off-state voltage. The gate loop is the exception to the lumped design: with the extracted gate network of the DPT board and $R_{g,on} = 2.5\ \Omega$ the turn-on reaches 49 A/ns and the high-side peak 105 V (90 V with the lumped gate loop). The DPT board is therefore

driven with 3.5 Ω (1.5 Ω shared per pair) and −1.9 V: the peak is 93 V with the shunts in the loop, the held-off gate stays at -0.5 V, 1.3 V below the minimum threshold, and the off-state ringing (about −3.4 V) inside the −4 V limit.

VI. Experimental Verification in the Final Paper

The final paper will add double-pulse measurements on the DPT board at 75 V up to the peak phase current of 322 A: the four cell currents through the coaxial shunts, v_{DS} and v_{GS} of every device through the MMCX jacks and spring-probe pads, the overshoot and gate bump with $R_{g,on}$ = 3.5 Ω and −1.9 V off-state, and the measured switching energies and their spread, compared with the simulation of Section V. Continuous operation of the load-test board will verify the thermal design.

VII. Conclusion

Because the phase current of an SPB is independent of the number of submodules, the device and the submodule voltage can be chosen for minimum loss and the available overshoot margin. For a 1200 V, 300 kW inverter this gives 16 submodules of 75 V with four 100 V GaN transistors per switch, driven in pairs by one dual-channel isolated driver each. Q3D extraction of the routed board gives 0.41 nH per cell with negligible coupling between the cells, and a double-pulse simulation with the full extracted network confirms equal sharing among the four cells and a peak voltage of 86 V with $R_{g,on}$ = 3 Ω. The gate loop of the centre driver, equal for all four paralleled devices but about ten times the assumed value, is the main deviation; the DPT board stays at 93 V with 3.5 Ω and −1.9 V, and a driver stage next to each pair is planned for the next revision.

VIII. References

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